

# Worked examples

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This vignette catalogs the loan types `mortgagemath` reproduces to the cent against published sources, organized by country convention. Each example shows: a one-paragraph **scenario**, the exact `LoanParams` configuration (sometimes via a convenience constructor), the equivalent `mortgagemath` CLI invocation, a live Python chunk emitting the schedule and the published anchors it matches, and a citation linking to both the source document and the fixture file.

For the broader institutional and mathematical history see the *History* vignette; for the validated worked-example matrix see *Validation*; for a 60-second orientation see *At a glance*.

## United States

### 30-year fixed-rate residential (CFPB H-25(B))

**Scenario.** The Consumer Financial Protection Bureau’s sample TILA-RESPA Integrated Disclosure form H-25(B), a fictional “Ficus Bank” closing-disclosure example. The published monthly P&I of \$761.78 implies `ROUND_HALF_UP` rounding; the unrounded closed-form value is \$761.7840..., which would round up to \$761.79 under the library’s default `ROUND_UP`.

```
mortgagemath payment --principal 162000 --rate 3.875 --term-months 360 \
--payment-rounding ROUND_HALF_UP --interest-rounding ROUND_HALF_UP
```

```
from decimal import Decimal
from mortgagemath import (
```

```

LoanParams,
PaymentRounding,
amortization_schedule,
canada_fixed_j2,
fixed_payment_mortgage,
periodic_payment,
us_15_year_fixed,
us_30_year_fixed,
us_actual_360_commercial,
)

cfpb = us_30_year_fixed(
    "162000.00",
    "3.875",
    payment_rounding=PaymentRounding.ROUND_HALF_UP,
)
print(f"Monthly P&I: ${periodic_payment(cfpb)} (CFPB H-25(B): $761.78)")

```

Monthly P&I: \$761.78 (CFPB H-25(B): \$761.78)

**Source.** CFPB Closing Disclosure Sample H-25(B) (2014). Fixture: cfpb\_h25b\_ficus\_30yr\_3875\_162000.{toml,csv}.

### 15-year fixed-rate residential (OpenStax, ROUND\_UP)

**Scenario.** OpenStax *Contemporary Mathematics* §6.8 home-loan example. The textbook explicitly states “*payment to lenders is always rounded up to the next penny*” — confirming the library’s default ROUND\_UP convention used by most US residential lenders.

```

openstax = us_15_year_fixed(
    "136700.00",
    "5.75",
    # ROUND_UP is the default; shown explicitly for clarity.
    payment_rounding=PaymentRounding.ROUND_UP,
    interest_rounding=PaymentRounding.ROUND_HALF_UP,
)
print(f"Monthly P&I: ${periodic_payment(openstax)} (OpenStax: $1,135.18)")

```

Monthly P&I: \$1135.18 (OpenStax: \$1,135.18)

**Source.** OpenStax *Contemporary Mathematics* §6.8, CC-BY-4.0. Fixture: openstax\_contemp\_home\_180mo\_575\_136700.{toml,csv}.

### Commercial Actual/360 with balloon (Fannie Mae §1103)

**Scenario.** Fannie Mae *Multifamily Selling and Servicing Guide* §1103, the Tier 2 SARM worked example. \$25 M / 5.5% / 10-year term on a 30-year amortization basis with an Actual/360 day-count convention. The borrower makes 120 monthly payments at the closed-form level, then owes a published balloon of \$20,885,505.83 at term.

```

mortgagemath schedule --principal 25000000 --rate 5.5 --term-months 120 \
    --amortization-period-months 360 --day-count actual/360 \
    --start-date 2018-12-01 \
    --payment-rounding ROUND_HALF_UP --interest-rounding ROUND_HALF_UP \
    --format csv > fnma_1103.csv

```

```

from datetime import date

fnma = us_actual_360_commercial(

```

```

    "25000000.00",
    "5.5",
    term_years=10,
    amortization_years=30,
    start_date=date(2018, 12, 1),
)
sched = amortization_schedule(fnma)
print(f"Monthly P&I:                ${periodic_payment(fnma):,}")
print(f"Balloon at term-120:        ${sched[120].balance:,}")
print(f" (Fannie Mae §1103 published: $141,947.25 / $20,885,505.83)")

```

```

Monthly P&I:                $141,947.25
Balloon at term-120:        $20,885,505.83
(Fannie Mae §1103 published: $141,947.25 / $20,885,505.83)

```

**Source.** Fannie Mae Multifamily Guide §1103 (eff. 2026-04-03). Fixture: fanniemaef\_mf\_1103\_25m\_550\_360mo.{toml,csv}.

## Adjustable-rate mortgage (Reg Z H-14)

**Scenario.** 12 CFR Part 1026 Appendix H Sample H-14 — the Federal-Reserve-promulgated regulatory ARM disclosure. \$10,000 / 30-year term / fully-amortizing 1/1 ARM at 1-year CMT + 3 pp margin, 2 pp annual periodic cap, 5 pp symmetric lifetime cap. The published example traces 15 years of historical adjustments (1982–1996) using actual 1-year CMT values, with the lifetime floor at 12.41% binding from 1986 onward.

| Yr | Year          | CMT index | Fully indexed | Cap binds?       | Effective rate |
|----|---------------|-----------|---------------|------------------|----------------|
| 1  | 1982          | 14.41%    | 17.41%        | (initial)        | 17.41%         |
| 2  | 1983          | 9.78%     | 12.78%        | periodic (down)  | 15.41%         |
| 3  | 1984          | 12.17%    | 15.17%        | —                | 15.17%         |
| 4  | 1985          | 7.66%     | 10.66%        | periodic (down)  | 13.17%         |
| 5  | 1986          | 6.36%     | 9.36%         | lifetime (floor) | 12.41%         |
| 6+ | 1987–<br>1996 | varies    | ≤ 9.36%       | lifetime holds   | 12.41%         |

```

mortgagemath schedule --principal 10000 --rate 17.41 --term-months 360 \
  --payment-rounding ROUND_HALF_UP --interest-rounding ROUND_HALF_UP \
  --rate-change 13:15.41 --rate-change 25:15.17 --rate-change 37:13.17 \
  --rate-change 49:12.41 # ... and similarly for years 6-15

```

```

from mortgagemath import RateChange, BalanceTracking

rates = ("15.41", "15.17", "13.17", "12.41", *(["12.41"] * 10))

regz_h14 = LoanParams(
    principal=Decimal("10000"),
    annual_rate=Decimal("17.41"),
    term_months=360,
    payment_rounding=PaymentRounding.ROUND_HALF_UP,
    interest_rounding=PaymentRounding.ROUND_HALF_UP,
    balance_tracking=BalanceTracking.ROUND_EACH,
    rate_schedule=tuple(
        RateChange(
            effective_payment_number=12 * (yr - 1) + 1,
            new_annual_rate=Decimal(rate),

```

```

    )
    for yr, rate in enumerate(rates, start=2)
    ),
)
sched = amortization_schedule(regz_h14)

```

| Anchor                          | Library    | Published | Match |
|---------------------------------|------------|-----------|-------|
| first row, year 1 (pmt 1)       | \$145.90   | \$145.90  | ✓     |
| balance after year 1 (pmt 12)   | \$9,989.37 | \$9989.37 | ✓     |
| first row, year 2 (pmt 13)      | \$129.81   | \$129.81  | ✓     |
| first row, year 5 (pmt 49)      | \$106.73   | \$106.73  | ✓     |
| balance after year 5 (pmt 60)   | \$9,848.94 | \$9848.94 | ✓     |
| balance after year 15 (pmt 180) | \$8,700.37 | \$8700.37 | ✓     |

**Source.** 12 CFR Part 1026 Appendix H Sample H-14. Fixture: regz\_apph\_h14\_arm\_10k\_1741\_360mo.{toml,csv}.

### ARM with payment cap + negative amortization (ProEducate)

**Scenario.** A *payment cap* is distinct from a *rate cap*. A rate cap bounds the interest rate; a payment cap bounds the dollar payment directly, regardless of what the new rate would otherwise produce. When the cap binds *and* the new periodic interest exceeds the capped payment, the unpaid interest is capitalized into the balance — the loan goes into negative amortization. ProEducate publishes a worked \$65,000 / 10% rising to 12% / 30-year / 7.5% annual cap example; the library reproduces every cell including the explicit \$420.90 of year-2 cumulative negative amortization.

```

mortgagemath schedule --principal 65000 --rate 10 --term-months 360 \
  --payment-rounding ROUND_HALF_UP --interest-rounding ROUND_HALF_UP \
  --rate-change 13:12:cap=1.075 --format csv

```

```

proeducate = LoanParams(
    principal=Decimal("65000"),
    annual_rate=Decimal("10"),
    term_months=360,
    payment_rounding=PaymentRounding.ROUND_HALF_UP,
    interest_rounding=PaymentRounding.ROUND_HALF_UP,
    balance_tracking=BalanceTracking.ROUND_EACH,
    rate_schedule=(
        RateChange(
            effective_payment_number=13,
            new_annual_rate=Decimal("12"),
            payment_cap_factor=Decimal("1.075"), # 7.5% annual cap
        ),
    ),
)
sched = amortization_schedule(proeducate)

```

- Year 1 P&I (closed-form at 10%): **\$570.42** (published \$570.42)
- Balance after pmt 12: **\$64,638.72** (published \$64,638.72)
- Year 2 capped P&I (= year-1 × 1.075): **\$613.20** (published \$613.20)
- Year 2 monthly principal (negative): **-\$33.19** (published -\$33.19)
- Cumulative neg-am over year 2: **\$420.90** (published \$420.90)
- Balance after pmt 24: **\$65,059.62** (published \$65,059.62)

**Source.** ProEducate *ARM Payment Caps*, 2014. Fixture: proeducate\_arm\_pmt\_cap\_65k\_10pct\_to\_12pct\_360mo.{toml,csv}.

## Carry-precision graduate-CRE (Geltner Ch 20)

**Scenario.** Geltner et al., *Commercial Real Estate Analysis and Investments*, online supplement Chapter 20 Exhibit 20-6 — a \$1 M / 12% / 30-year constant-payment mortgage. The textbook uses Excel-default carry-precision balance tracking throughout, so the library's CARRY\_PRECISION mode is the right configuration.

```
geltner = LoanParams(  
    principal=Decimal("1000000.00"),  
    annual_rate=Decimal("12"),  
    term_months=360,  
    payment_rounding=PaymentRounding.ROUND_HALF_UP,  
    interest_rounding=PaymentRounding.ROUND_HALF_UP,  
    balance_tracking=BalanceTracking.CARRY_PRECISION,  
)  
print(f"Monthly P&I: ${periodic_payment(geltner):.2f} (Geltner: $10,286.13)")
```

Monthly P&I: \$10,286.13 (Geltner: \$10,286.13)

**Source.** Geltner, Miller, van de Minne, Eichholtz, Lindenthal, & Shen, *Commercial Real Estate Analysis and Investments*, Routledge, 2024. Fixture: geltner\_ch20\_cpm\_1m\_1200\_360mo.{toml,csv}.

## Half-cent rounding boundary (synthetic)

**Scenario.** Three paired synthetic fixtures engineered so month-1 unrounded interest equals exactly \$400.005 — a half-cent boundary that distinguishes ROUND\_HALF\_UP from ROUND\_HALF\_EVEN and the closed-form payment from ROUND\_UP. \$100,001.25 / 4.80% / 30-year. Same loan; three rounding configurations.

```
synth_principal = Decimal("100001.25")  
synth_rate = Decimal("4.80")  
  
for rounding, expected in [  
    (PaymentRounding.ROUND_HALF_UP, "524.67"),  
    (PaymentRounding.ROUND_UP, "524.68"),  
]:  
    loan = LoanParams(  
        principal=synth_principal,  
        annual_rate=synth_rate,  
        term_months=360,  
        payment_rounding=rounding,  
        interest_rounding=PaymentRounding.ROUND_HALF_UP,  
    )  
    print(f"{rounding.value:14s} → ${periodic_payment(loan)} " +  
          f"(expected ${expected})")
```

ROUND\_HALF\_UP → \$524.67 (expected \$524.67)  
ROUND\_UP → \$524.68 (expected \$524.68)

**Sources.** Synthetic boundary fixtures. synthetic\_halfcent\_halfup\_360\_480\_100001p25.{toml,csv} and companions.

## Effective-annual on monthly cadence (Skinner 1913 piano)

**Scenario.** Skinner's 1913 *Mathematical Theory of Investment* §42 Example 3 — a piano costing \$500 to be paid off at 6% *effective annual* over 5 years monthly. The actuarial convention treats 6% as the effective-annual rate and derives the equivalent nominal-monthly rate  $i^{(12)} = 12 \cdot \left( (1.06)^{1/12} - 1 \right) \approx 0.0584$ , *not*  $6/12 = 0.5\%$ . Naïve Compounding.MONTHLY gives \$9.67 — a 4-cent miss. The library's Compounding.ANNUAL mode reproduces Skinner's published \$9.63 exactly.

```
from mortgagemath import Compounding, PaymentFrequency
```

```

skinner_piano = LoanParams(
    principal=Decimal("500.00"),
    annual_rate=Decimal("6"),
    term_months=60,
    payment_frequency=PaymentFrequency.MONTHLY,
    compounding=Compounding.ANNUAL,
    payment_rounding=PaymentRounding.ROUND_HALF_UP,
    interest_rounding=PaymentRounding.ROUND_HALF_UP,
)
print(f"Monthly piano payment: ${periodic_payment(skinner_piano)} "
      f"(Skinner §42 Ex 3: $9.63)")

```

Monthly piano payment: \$9.63 (Skinner §42 Ex 3: \$9.63)

**Source.** Skinner, *The Mathematical Theory of Investment*, Boston: Ginn and Company, 1913. Public domain on Internet Archive. Fixture: `skinner_1913_42_ex3_500_6pct_5yr_monthly.{toml,csv}`.

## Annual cadence + annual compounding (Arcones SOA FM)

**Scenario.** Miguel Arcones’s *Manual for SOA Exam FM / CAS Exam 2*, §4.1 Example 4 — \$20,000 / 8% effective annual / 12 annual payments. The full 12-row schedule is published in the source; the library reproduces every cell exactly under `Compounding.ANNUAL + PaymentFrequency.ANNUAL`.

```

arcones = LoanParams(
    principal=Decimal("20000.00"),
    annual_rate=Decimal("8"),
    term_months=144,
    payment_frequency=PaymentFrequency.ANNUAL,
    compounding=Compounding.ANNUAL,
    payment_rounding=PaymentRounding.ROUND_HALF_UP,
    interest_rounding=PaymentRounding.ROUND_HALF_UP,
    balance_tracking=BalanceTracking.ROUND_EACH,
)
sched = amortization_schedule(arcones)
print(f"Annual installment:      ${periodic_payment(arcones):.2f}")
print(f"Final-row trueup (yr 12): ${sched[12].payment} (Arcones: $2,653.91)")

```

Annual installment: \$2,653.90  
 Final-row trueup (yr 12): \$2653.91 (Arcones: \$2,653.91)

**Source.** Arcones, *Manual for SOA Exam FM / CAS Exam 2* §4.1, 2009. Fixture: `arcones_soa_fm_4_1_ex4_20k_8pct_12yr_annual.{toml,csv}`.

## Given-payment, find-term (FHLBB 1935 Plan A) — new in v0.6.0

**Scenario.** The Federal Home Loan Bank Board’s *FHLBB Review* of March 1935 published the earliest U.S. federal-authority worked direct-reduction amortization schedule, in an article recommending the direct-reduction plan over the dominant share-accumulation B&L scheme. Plan A: \$3,000 at 6% with *monthly* interest credit, payment chosen by convention as 1% of original principal = \$30.00 per month. The schedule retires in 138 full payments of \$30 plus a 139th payment of \$29.27.

This is the historical “given-payment, find-term” convention: the lender chose a round payment (typically a percentage of original principal) and accepted whatever final-payment trueup the math produced. The library’s default closed-form mode (term in, payment out) cannot reproduce this — it derives a level \$29.9964 for term=139, which rounds to \$30.00 with a \$29.35 final-row trueup. The published \$29.27 is the carry-precision balance after 138 payments  $\times (1 + r)$ , rounded once.

v0.6.0’s `payment_override` field pins the payment and applies a “round-the-total” final-row trueup, reproducing every published cell.

```
mortgagemath schedule --principal 3000 --rate 6 --term-months 139 \
--payment-rounding ROUND_HALF_UP --interest-rounding ROUND_HALF_UP \
--balance-tracking carry_precision \
--payment-override 30 --format table
```

```
fhlbb = fixed_payment_mortgage(
    "3000.00",
    "6",
    "30.00",
    term_months=139,
)
sched = amortization_schedule(fhlbb)
for n in (1, 138, 139):
    inst = sched[n]
    print(f"month {n:>3}: payment=${inst.payment} "
          f"interest=${inst.interest} principal=${inst.principal} "
          f"balance=${inst.balance}")
print(f"\nFinal payment: ${sched[139].payment} (FHLBB published $29.27)")
```

```
month 1: payment=$30.00 interest=$15.00 principal=$15.00 balance=$2985.00
month 138: payment=$30.00 interest=$0.29 principal=$29.71 balance=$29.13
month 139: payment=$29.27 interest=$0.15 principal=$29.12 balance=$0

Final payment: $29.27 (FHLBB published $29.27)
```

**Source.** FHLBB *Federal Home Loan Bank Review*, Vol. 1 No. 6, March 1935, pp. 187–198, public domain. Fixture: fhlbb\_1935\_plan\_a\_3k\_6pct\_30\_per\_mo.{toml,csv}.

## Canada

### Semi-annual $j_2$ with monthly payments (Olivier Chans)

**Scenario.** Canadian residential mortgages quote rates as *semi-annually compounded* ( $j_2$ ), per Section 6 of the federal *Interest Act*. A US “5% / 30 yr” loan and a Canadian “5% / 30 yr” loan have different periodic rates and different monthly payments even though both quote 5%. Olivier’s *Business Math* §13.4 walks through the Chans family’s first term: \$350,100 /  $j_2 = 4.9\%$  / 3-year fixed term on a 20-year amortization basis, monthly payments. At end of the 3-year term the unpaid balance is the balloon to be renewed at then-prevailing rates.

#### Note

**US:** monthly periodic rate is  $r/12$  directly. A 5% loan uses  $5/1200 = 0.004167$  per month.

**Canadian ( $j_2$ ):** the equivalent monthly periodic rate is  $(1 + j_2/200)^{2/12} - 1$ . For  $j_2 = 5\%$ , this is  $(1.025)^{1/6} - 1 \approx 0.41239\%$ , not  $5/12 \approx 0.4167\%$ .

```
mortgagemath payment --principal 350100 --rate 4.9 --term-months 36 \
--amortization-period-months 240 --compounding semi_annual \
--payment-rounding ROUND_HALF_UP --interest-rounding ROUND_HALF_UP
```

```
chans = canada_fixed_j2("350100", "4.9", amortization_years=20, term_years=3)
print(f"Monthly P&I: {periodic_payment(chans):,}")
sched = amortization_schedule(chans)
print(f"Balance after pmt 36 (renewal balloon): ${sched[36].balance:,}")
print(f" (Olivier published: $2,281.73 / $316,593.49)")
```

```
Monthly P&I:                $2,281.73
Balance after pmt 36 (renewal balloon): $316,593.49
(Olivier published: $2,281.73 / $316,593.49)
```

**Source.** Olivier, *Business Math: A Step-by-Step Handbook* §13.4, LibreTexts (CC-BY-NC-SA), 2021. Fixture: `olivier_chans_350100_490_36mo_term_240mo.{toml,csv}` plus the companion `olivier_chans_renewal_316593p49_585_204mo.{toml,csv}` for the renewal-term scenario at 5.85%.

### Semi-annual $j_2$ with quarterly payments (eCampus §4.4.1)

**Scenario.** Canadian semi-annual compounding combines naturally with non-monthly payment cadences. eCampus Ontario's *Mathematics of Finance* §4.4.1 publishes a **quarterly-payment** worked example:  $\$297,500 / j_2 = 3.8\% / 3\text{-year fixed term}$  on a 20-year amortization, quarterly payments.

```
ecampus_q = canada_fixed_j2(
    "297500",
    "3.8",
    amortization_years=20,
    term_years=3,
    payment_frequency=PaymentFrequency.QUARTERLY,
)
sched = amortization_schedule(ecampus_q)
print(f"Quarterly P&I:           ${periodic_payment(ecampus_q):,}")
print(f"Balance after 12 quarters:  ${sched[12].balance:,}")
print(f" (eCampus published: $5,317.62 / $265,830.61)")
```

```
Quarterly P&I:                $5,317.62
Balance after 12 quarters:    $265,830.61
(eCampus published: $5,317.62 / $265,830.61)
```

The schedule walks 12 quarterly rows (not 12 monthly), because  $\text{term\_months} \cdot \text{payments\_per\_year} / 12 = 36 \cdot 4 / 12 = 12$  payments.

**Source.** eCampus Ontario *Mathematics of Finance* §4.4.1, CC-BY-4.0. Fixture: `ecampus_finmath_4_4_1_297500_380_q_36mo_term_240mo.{toml,csv}` plus the renewal companion at  $j_2 = 2.5\%$ .

### Monthly compounding on Canadian textbook examples

The same eCampus textbook also publishes US-convention examples (monthly compounding, monthly payments) — these appear in §4.3 Examples 4.3.1 (Pearline,  $\$10,000 / 10\%$  effective annual / 4-year annual, full schedule), Exercise 2 (Erika,  $\$32,600 / 4.83\% / 9\text{-year monthly}$  with year-aggregate anchors), and Exercise 3 (Johnetta,  $\$20,200 / 3.53\% / 8\text{-year monthly}$  with a mid-schedule probe at payment 60). All three reproduce to the cent under the library's defaults; see the *Validation* vignette for the per-fixture matrix.

#### Note

**Why this matters.** Without Compounding.SEMI\_ANNUAL, a Canadian-style mortgage calculator written in stdlib Python will quote a payment about 0.7% high on a typical 25-year loan — a few dollars per month, but several thousand dollars over the loan life. Most Python amortization libraries silently get this wrong.

## France

### MoneyVox tableau d'amortissement (12-month monthly)

**Scenario.** A worked example from MoneyVox, one of France's largest personal finance sites. €10,000 at 5% over 12 monthly payments. The source publishes a full *tableau d'amortissement* with separate columns for *capital amorti*, *intérêts*, and *assurance emprunteur* (0.35% on initial capital = €2.92/month). This fixture validates the full published total mensualité: the pure P+I schedule (hors assurance) plus the flat insurance loading. All 12 rows match the library cell-for-cell.



```

fr_loan = LoanParams(
    principal=Decimal("10000"),
    annual_rate=Decimal("5"),
    term_months=12,
    payment_rounding=PaymentRounding.ROUND_HALF_UP,
    interest_rounding=PaymentRounding.ROUND_HALF_UP,
    fee_per_period=Decimal("2.92"),
)
sched = amortization_schedule(fr_loan)
print(f"Mensualité hors assurance: €{sched[1].payment - sched[1].fee}")
print(f"Assurance: €{sched[1].fee}")
print(f" Month 1: intérêts €{sched[1].interest}, "
      f"capital amorti €{sched[1].principal}, "
      f"mensualité totale €{sched[1].payment}, "
      f"CRD €{sched[1].balance}")
print(f" Month 12: intérêts €{sched[12].interest}, "
      f"capital amorti €{sched[12].principal}, "
      f"mensualité totale €{sched[12].payment}, "
      f"CRD €{sched[12].balance}")

```

```

Mensualité hors assurance: €856.07
Assurance: €2.92
Month 1: intérêts €41.67, capital amorti €814.40, mensualité totale €858.99, CRD €9185.60
Month 12: intérêts €3.55, capital amorti €852.57, mensualité totale €859.04, CRD €0.00

```

`periodic_payment(fr_loan)` remains €856.07, the actuarially pure annuity. `amortization_schedule(fr_loan)` adds `fee_per_period` to each row's payment, so the row-1 gross mensualité is €858.99 and the final trueup row is €859.04.

**Source.** MoneyVox *Tableau d'amortissement*. Fixture: `moneyvox_fr_10k_5pct_12mo.{toml,csv}`.

The 1852 Crédit Foncier de France structure embeds an administrative loading into the published *annuité* — a structure now modeled by the same `fee_per_period` field. See *History* §10 for details and remaining historical source gaps.

## United Kingdom

The British **Benefit Building Societies Act 1836** provided the statutory frame for what would become the U.K.'s dominant retail mortgage lender through most of the twentieth century. From the 1990s onward, U.K. building-society direct-reduction mortgages converge with the U.S. convention; a U.K. fixture is mechanically a duplicate of an existing U.S. example. This section reserves the slot until a specific U.K. published table surfaces.

## Japan

### JHF Flat 35 (30-year fixed-rate, yen precision)

**Scenario.** The Japan Housing Finance Agency (国住宅金融公庫) publishes a comparison of repayment methods for its Flat 35 program. ¥20,000,000 at 1.5% fixed over 30 years using *ganri kinto hensai* (元金均等返済, equal total payment / standard annuity). The monthly payment is ¥69,024.

This fixture demonstrates the library's `currency_unit` feature: Japanese yen has no subunit, so all amounts are rounded to whole yen. Setting `currency_unit=Decimal("1")` instructs the library to quantize all monetary values to the nearest yen instead of the default cent.

```

jp_loan = LoanParams(
    principal=Decimal("20000000"),
    annual_rate=Decimal("1.5"),
    term_months=360,
    payment_rounding=PaymentRounding.ROUND_HALF_UP,
    interest_rounding=PaymentRounding.ROUND_HALF_UP,
    currency_unit=Decimal("1"),
)

```

```
)  
print(f"Monthly payment: ¥{periodic_payment(jp_loan),}")
```

Monthly payment: ¥69,024

**Source.** JHF Flat 35 repayment comparison. Fixture: `jhf_flat35_20m_150_360mo.{toml,csv}`.

## Australia

Victoria's **Credit Foncier Act 1896** established a state-owned mortgage bank on the CF model, and the term *credit foncier* persisted in Australian mortgage-banking parlance for decades thereafter. A 1900s–1950s state-bank schedule with fee loading would land here if a row-level published source surfaces.